EC 504
Fall, 2025
Homework 7

Due Tuesday, December 9, 2025, 11:59 pm

Problem 7.1 (10 pts)

For the KMP algorithm, compute the prefix function for the following string: "abracadabra".

Problem 7.2 (15 pts)

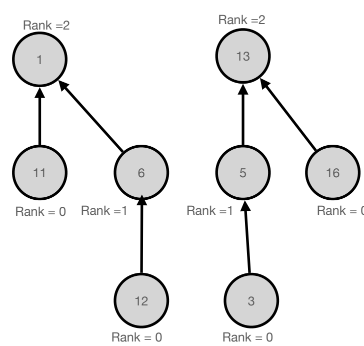
For the Aho-Corasick algorithm, compute the suffix trie that would be used to search for the following strings simultaneously:

```
"other"
"otter"
"often"
"threw"
"these"
```

You do not need to include output links, only prefix links.

Problem 7.3 (5 pts)

Old exam question: Consider the two trees, shown on the right, that are part of a disjoint set forest. Suppose we find a relation between keys 12 and 3. Show the disjoint set tree that results from the union operation using merge by rank and path compression: if two roots have the same rank, merge the larger value root under the smaller value root and upon find, compress the paths.



Problem 7.4 (15 pts)

Assume you have a scheduling problem with 10 customers. Customer i has value V_i, and requires processing time T_i. The values of V_i and T_i are listed below as arrays:

```
V = {3,4,7,5,2,3,5,8,9,6}
T = {2,3,4,3,1,3,3,5,6,4}
```

Assume that jobs can be scheduled at any start time, and there is a total amount of time T to allocate for the jobs. Hence, you can view the scheduling problem as a version of a knapsack problem.

Suppose that jobs must be scheduled fully to receive any value, so that no partial credit is given to incomplete jobs. Use dynamic programming to find the optimal value which can be scheduled in a total of 9, 10, 11, 12 time units. (Note: this may be easier to do in MATLAB or Python than by hand...)

Problem 7.5 (15 pts)

Consider the two words: *perplex* and *peruse*. Consider them as a sequence of letters. We would like to find the best alignment of the two letter sequences, with the following costs:

- Incur a penalty of 1 for any skipped letter.
- Incur a penalty of 3 for matching a consonant to a vowel.
- Incur a penalty of 2 for mismatching a consonant to another consonant or a vowel to another vowel.
- Incur a reward of 1 for correctly matching two letters.

Using this reward structure, find the minimum cost alignment. State the cost of the alignment.

Problem 7.6 (15 points) Consider the following pairs of coordinates: (7,2), (2,3), (4,6), (5,9), (8,1), (10,4), (13,5) (6,7), (9,8). Arrange them in a quadtree, using the maximum range 0-15 for each of the coordinates.

Problem 7.7 (10 points)

Consider the following pairs of coordinates: (7,2), (2,3), (4,6), (5,9), (8,1), (10,4), (13,5) (6,7), (9,8). Show the 2-d binary search tree which results from inserting these elements in the order given, with no attempt to balance them.

Problem 7.8 (15 points)

Consider the following pairs of coordinates: (7,2), (2,3), (4,6), (5,9), (8,1), (10,4), (13,5) (6,7), (9,8). Form a balanced 2-d binary search tree using the median of the elements to split. To make things unique, when you have to find a median of a list with an even number of entries, choose the smaller of the two entries as the median.